BLOOD VASCULAR SYSTEM, CAPILLARIES, ARTERIES, VEINS, ANASTOMOSIS

LEARNING OBJECTIVE

At the end of lecture, student should be able to know:

- Components of blood vascular system
- Recognize general structure of blood vessels.
- Capillaries and their types
- Recognize the structure of different types of arteries
- Recognize the structure fo different types of veins
- Discuss microcirculation, anastomosis, end-arteries

THE CARDIOVASCULAR SYSTEM

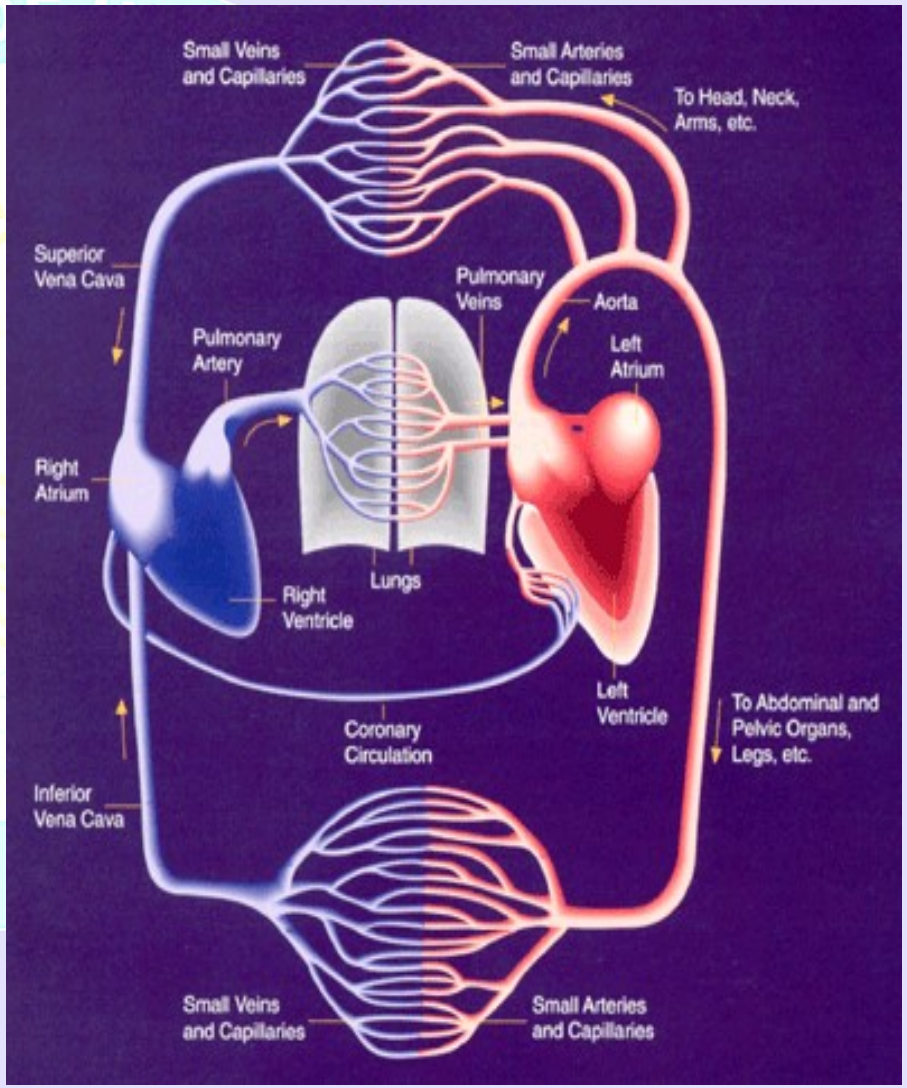
Concerned with

transport of blood &

lymph throughout the body.

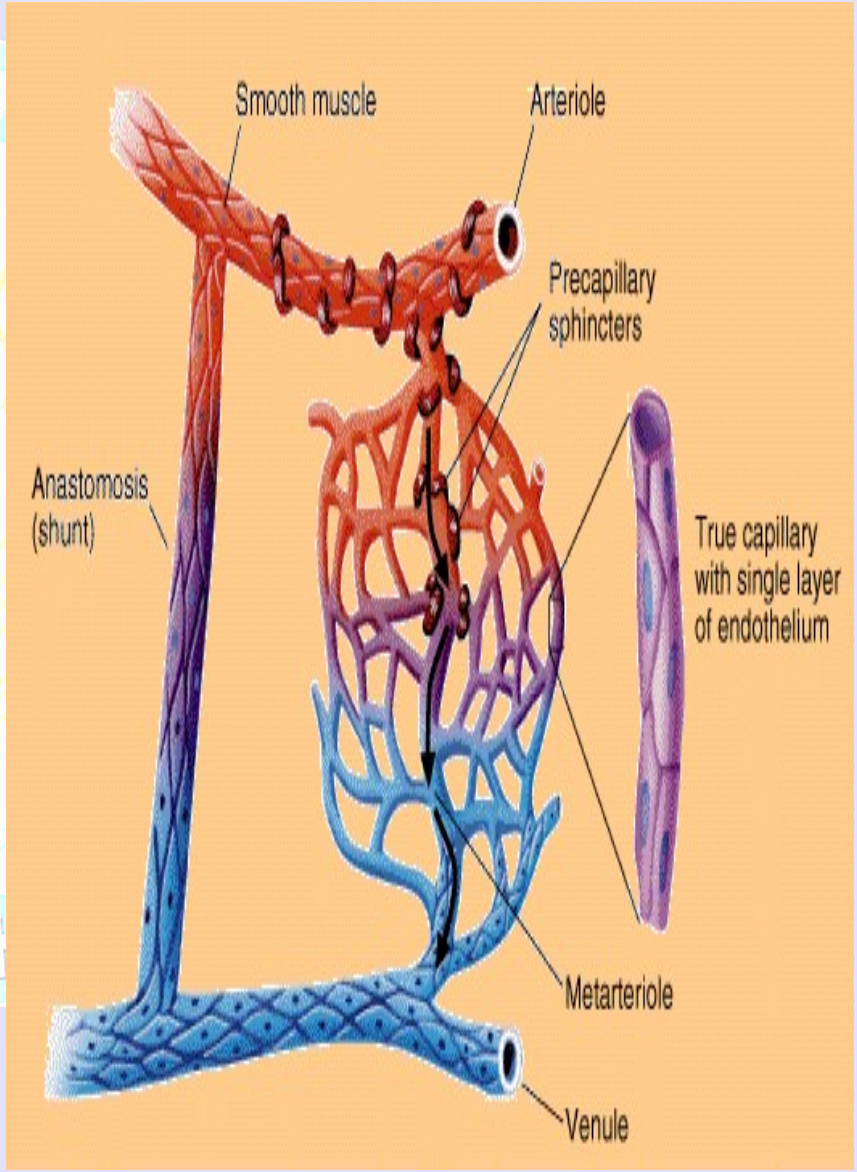
Divided into four major components:

- The heart
- The macrocirculation
- The microcirculation
- The lymph vascular system.



MACROCIRCULATION

- Comprises of all vessels, both arteries and veins
- The vessels supply and drain a network of fine vessels interposed between them, the capillaries.
- This network is also called the capillary bed.
- Water and other components of the blood plasma which exude from the blood vessels form the interstitial fluid, which is returned to the circulation by the lymph vascular system.



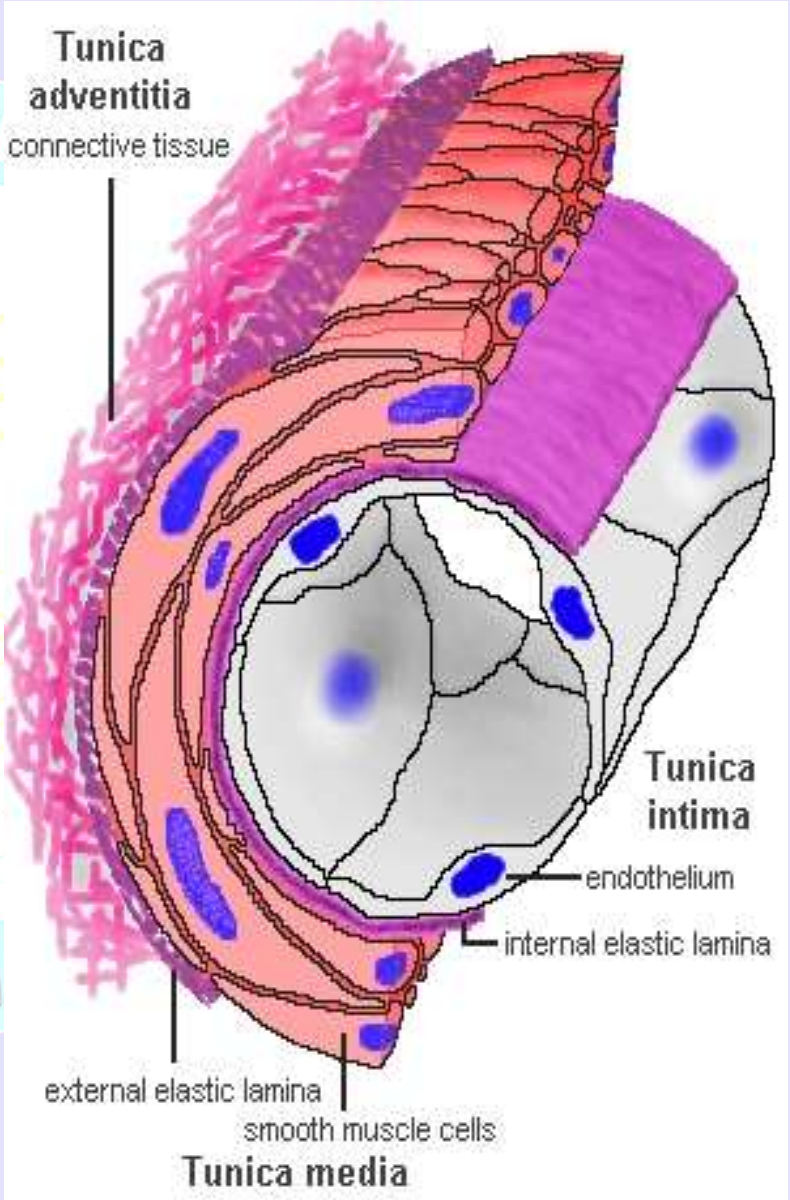
GENERAL STRUCTURE OF BLOOD VESSELS

- The histological appearances of vessels of different sizes (arterioles vs. arteries) and different types (arteries vs. veins) are different from each other.
- The division of the walls of the blood vessels into 3 layers or tunics:

➢ The tunica intima

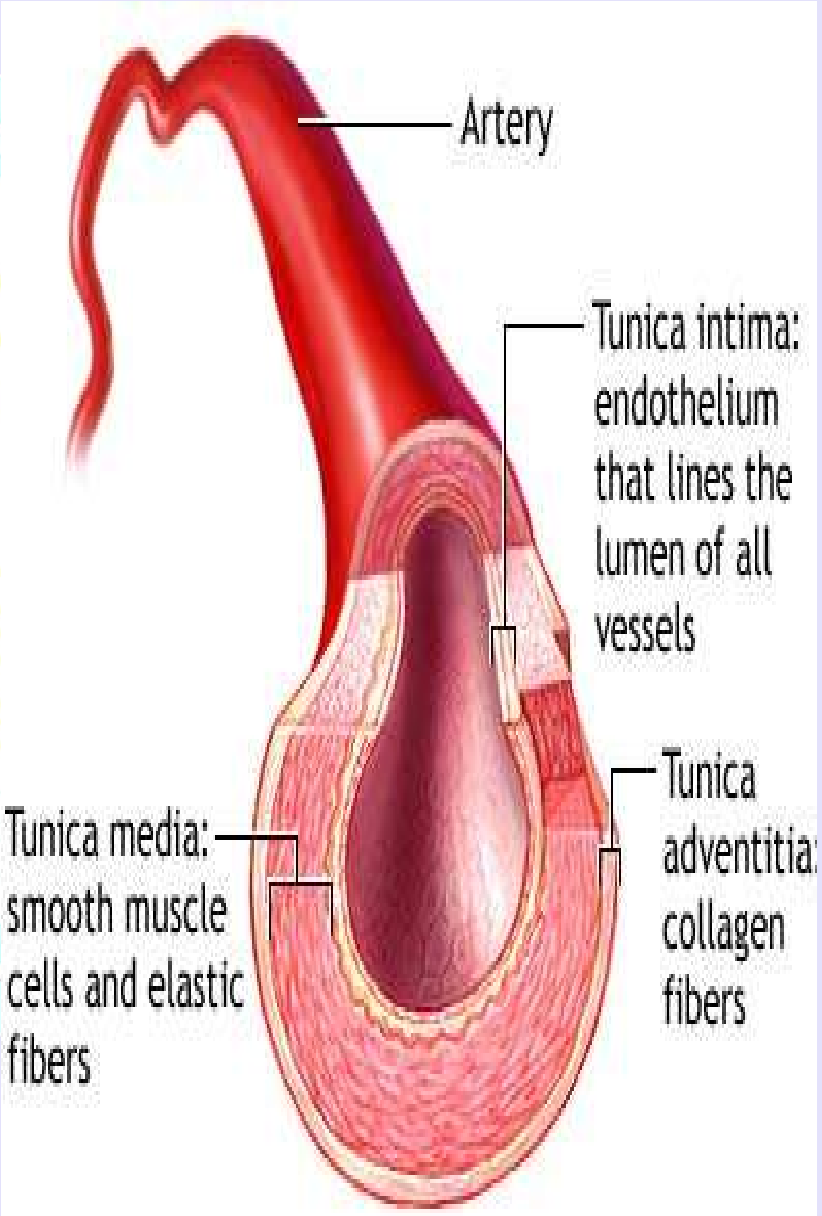
➢ The tunica media.

➢ The tunica adventitia



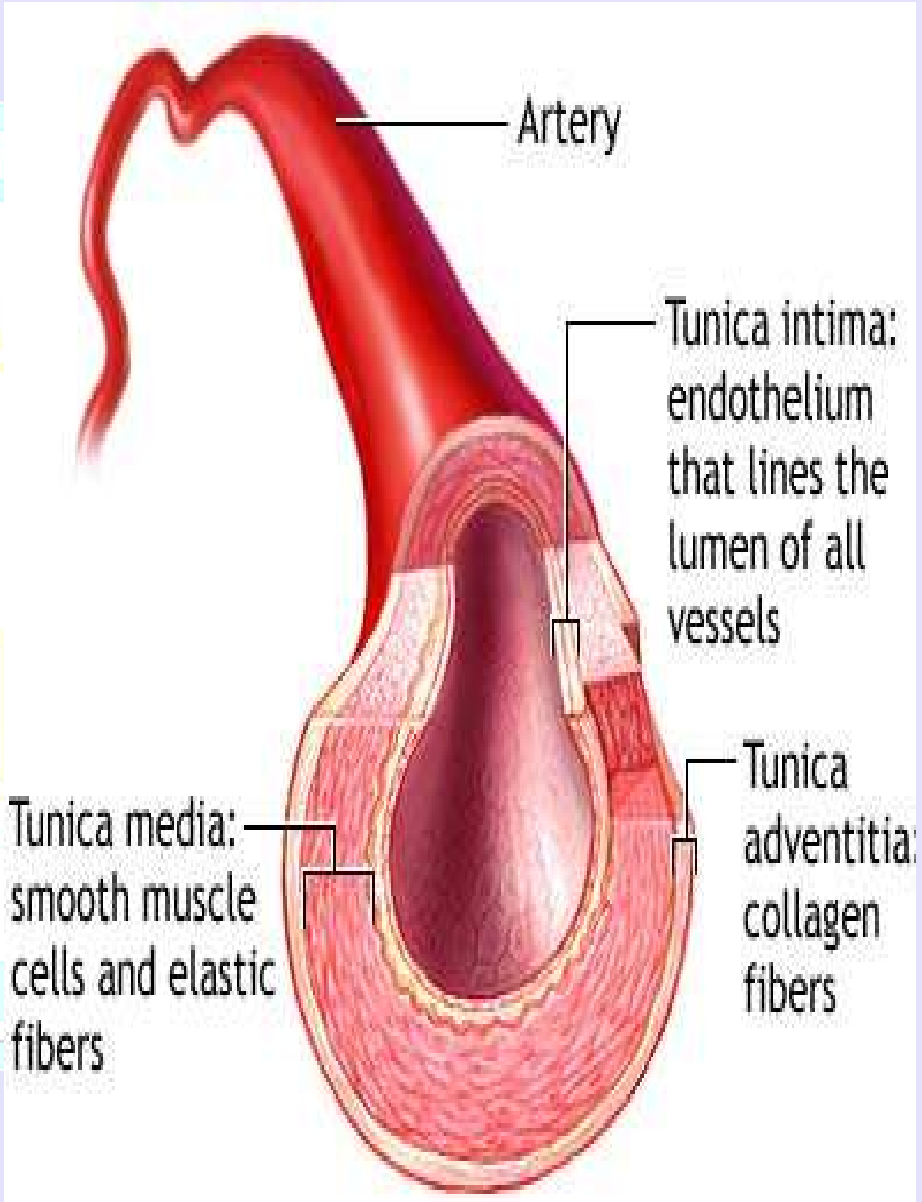
THE TUNICA INTIMA

- Delimits the vessel wall towards the lumen of the vessel and comprises of its endothelial lining (typically simple, squamous) and associated connective tissue.
- Beneath the connective tissue, we find the internal elastic lamina, which delimits the tunica intima from the tunica media.



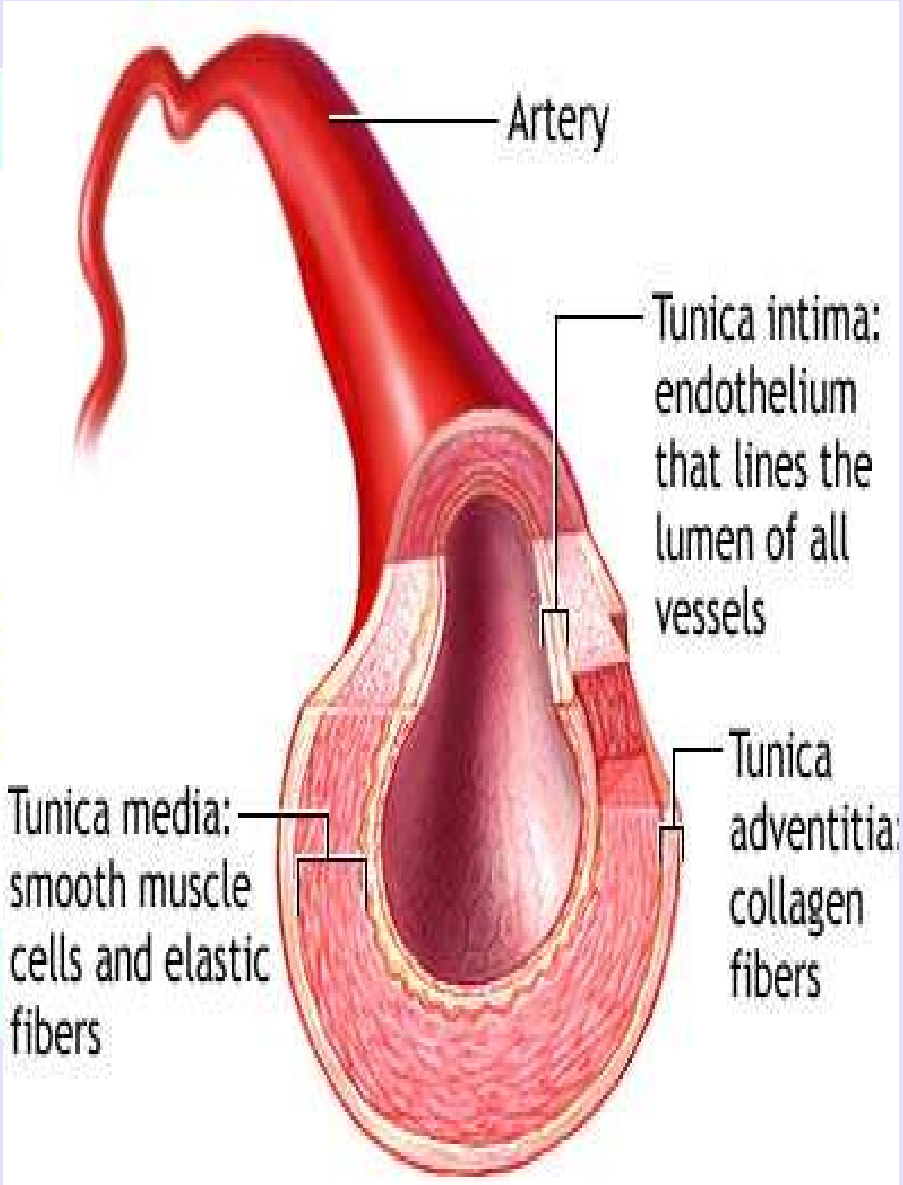
THE TUNICA MEDIA

- The tunica media is formed by a layer of circumferential smooth muscle and variable amounts of connective tissue.
- A second layer of elastic fibers, the external elastic lamina, is located beneath the smooth muscle.
- It delimits the tunica media from the tunica adventitia.



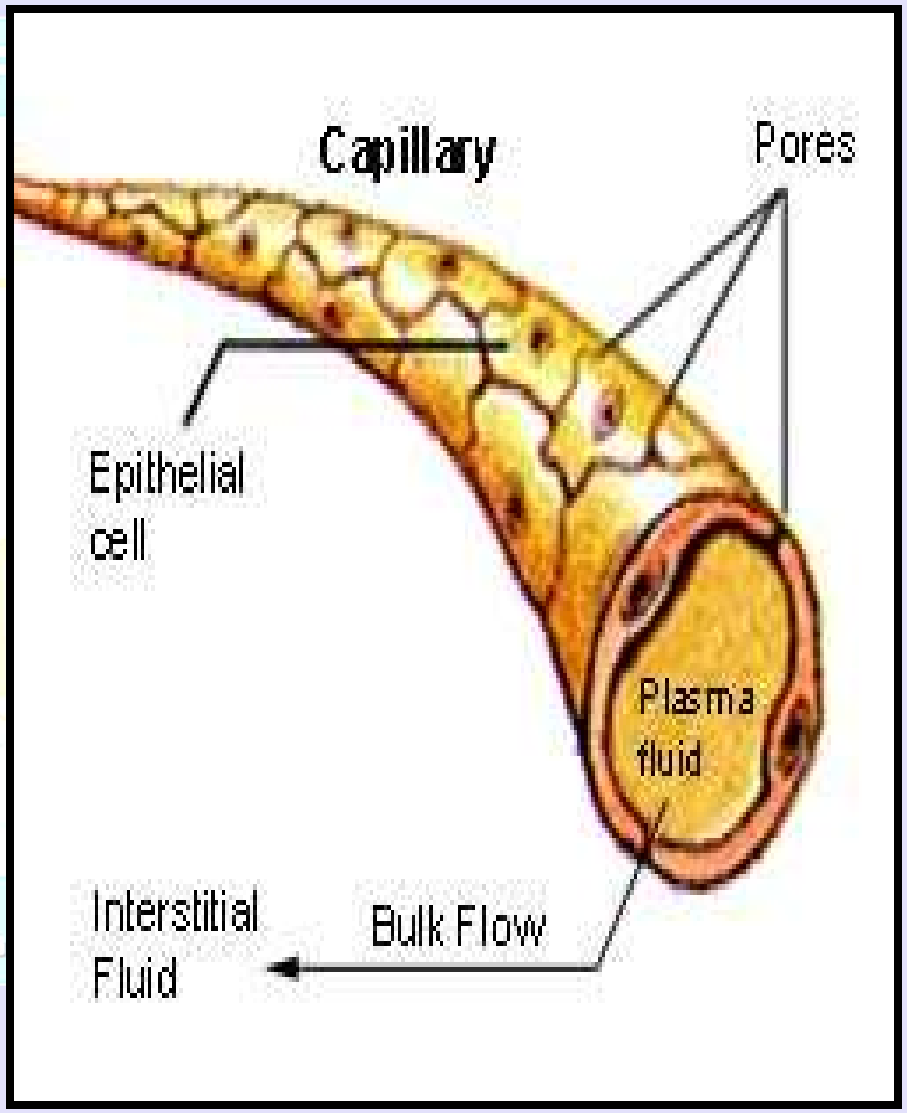
THE TUNICA ADVENTITIA

- Consist mainly of connective tissue fibres.
- The tunica adventitia blends with the connective tissue surrounding the vessel.
- The definition of the outer limit of the tunica adventitia is therefore somewhat arbitrary.



CAPILLARIES

- The sum of the diameters of all capillaries is significantly larger than that of the aorta which results in decreases in blood pressure and flow rate.
- Capillaries are very small vessels.
- Their diameter ranges from 7-9 μm.



CAPILLARIES

The low rate of blood flow and large surface area facilitate the functions of capillaries in

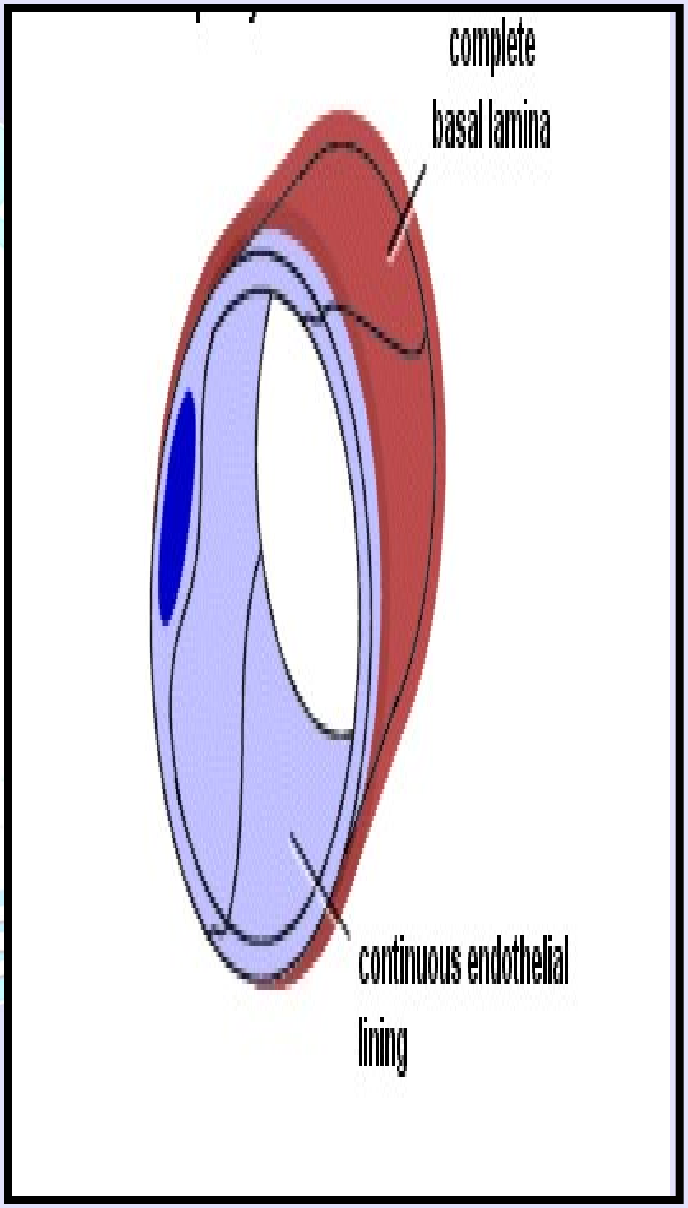
- providing nutrients and oxygen to the surrounding tissue
- In the absorption of nutrients, waste products and carbon dioxide
- In the excretion of waste products from the body.

- In capillaries, only the tunica intima is present, which typically only consists of the endothelium, its basal lamina and an incomplete layer of cells surrounding the capillary, the pericytes.
- Pericytes have contractile properties and can regulate blood flow in capillaries.
- Three types of capillaries can be distinguished based on features of the endothelium.

➢ Continuous capillaries
➢ Fenestrated capillaries
➢ Sinusoidal capillaries

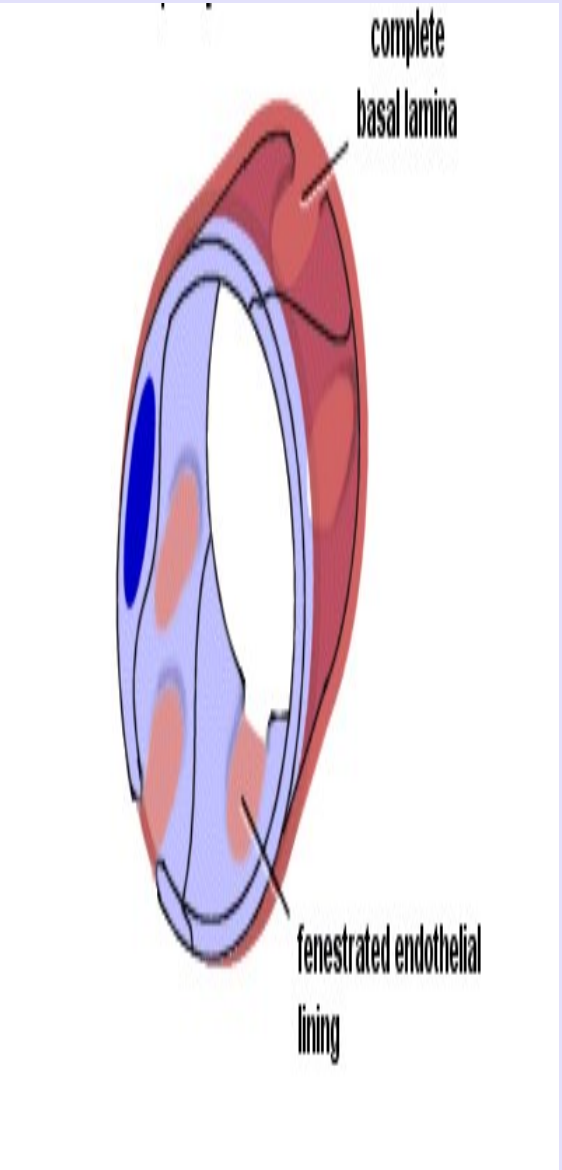
CONTINUOUS CAPILLARIES

- Are formed by "continuous" endothelial cells and basal lamina.
- The endothelial cell and the basal lamina do not form openings, which would allow substances to pass the capillary wall without passing through both the endothelial cell and the basal lamina.
- Both endothelial cells and the basal lamina can act as selective filters in continuous capillaries.



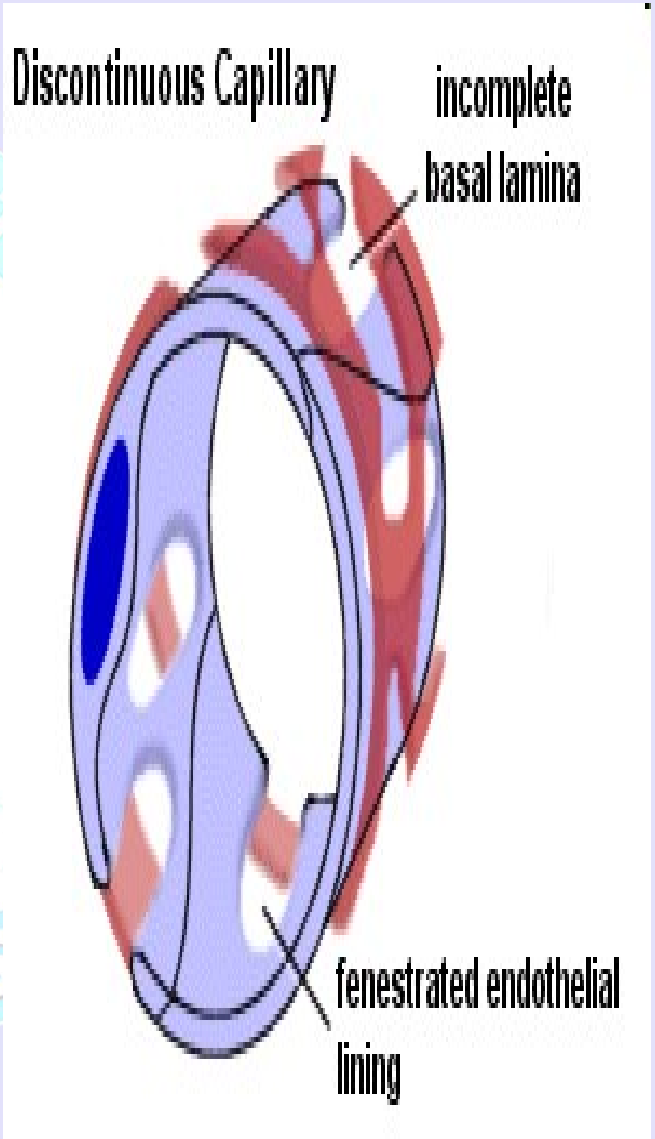
FENESTRATED CAPILLARIES

- The endothelial cell body form small openings called fenestrations, which allow blood components & interstitial fluid to bypass the endothelial cells on their way to or from the tissue surrounding the capillary.
- Fenestrations represent or arise from pinocytotic vesicles which open onto both the luminal and basal surfaces of the cell.
- The extent of the fenestration depend on the physiological state of the surrounding tissue.
- The endothelial cells are surrounded by a continuous basal lamina which can act as selective filter.



SINUSOIDAL CAPILLARIES

- Are formed by fenestrated endothelial cells, which may not form a complete layer of cells.
- The basal lamina is also incomplete.
- Sinusoidal capillaries form large irregularly shaped vessels, sinusoids.
- They are found where free exchange of substances or even cells between bloodstream and organ is advantageous.

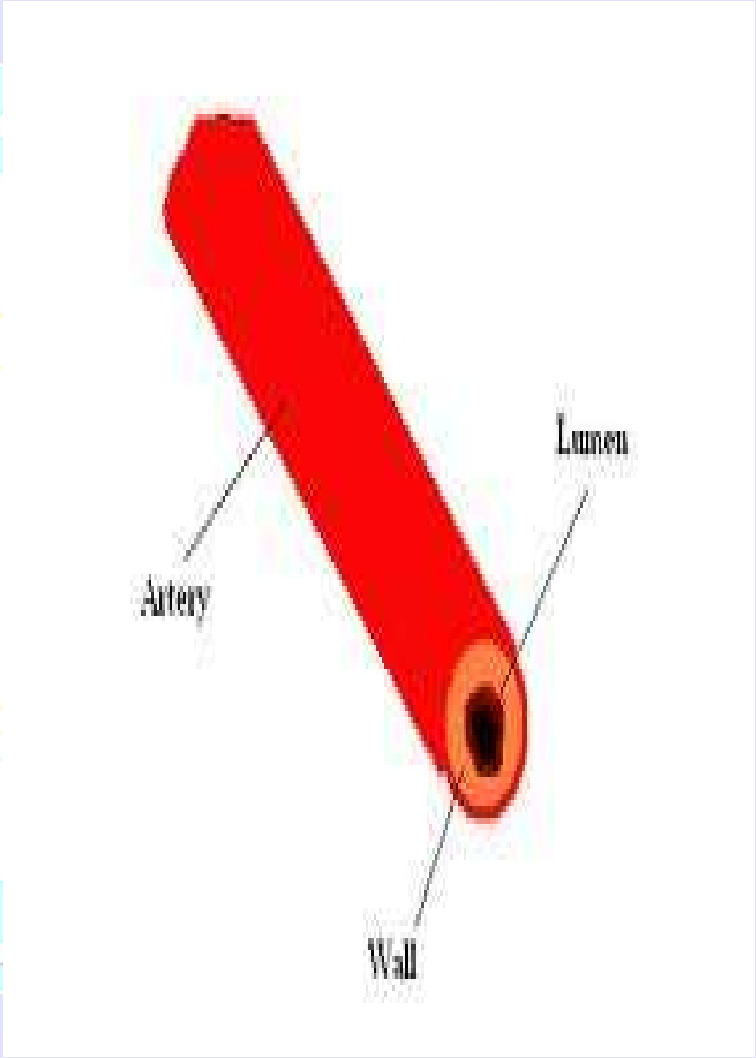


ARTERIES

Depending on their size, the arteries are classified into the following three main types:

- Arterioles and small arteries.
- Medium size arteries.
- Large arteries.

The structure and relative thickness of the three component tunics vary according to the type of the artery .

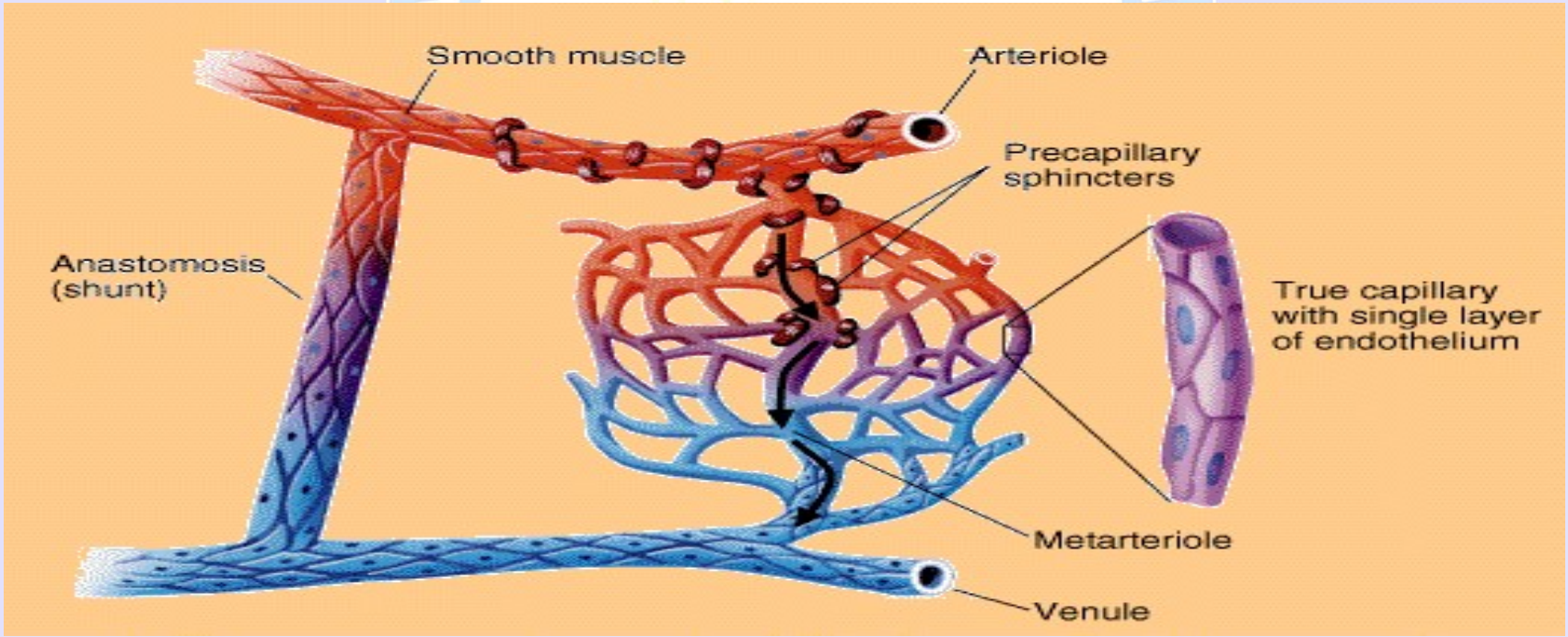


ARTERIOLES AND SMALL ARTERIES

- All three tunics are distinguishable.
- The intima consists only of endothelium.
- Subendotheliallayer is very thin.
- Internal elastic lamina may be seen in the small arteries.
- The media is composed of one or two circularly arranged layers of smooth muscle cells.
- The adventitia consists of a thin layer of longitudinal collagenous and elastic fibers.

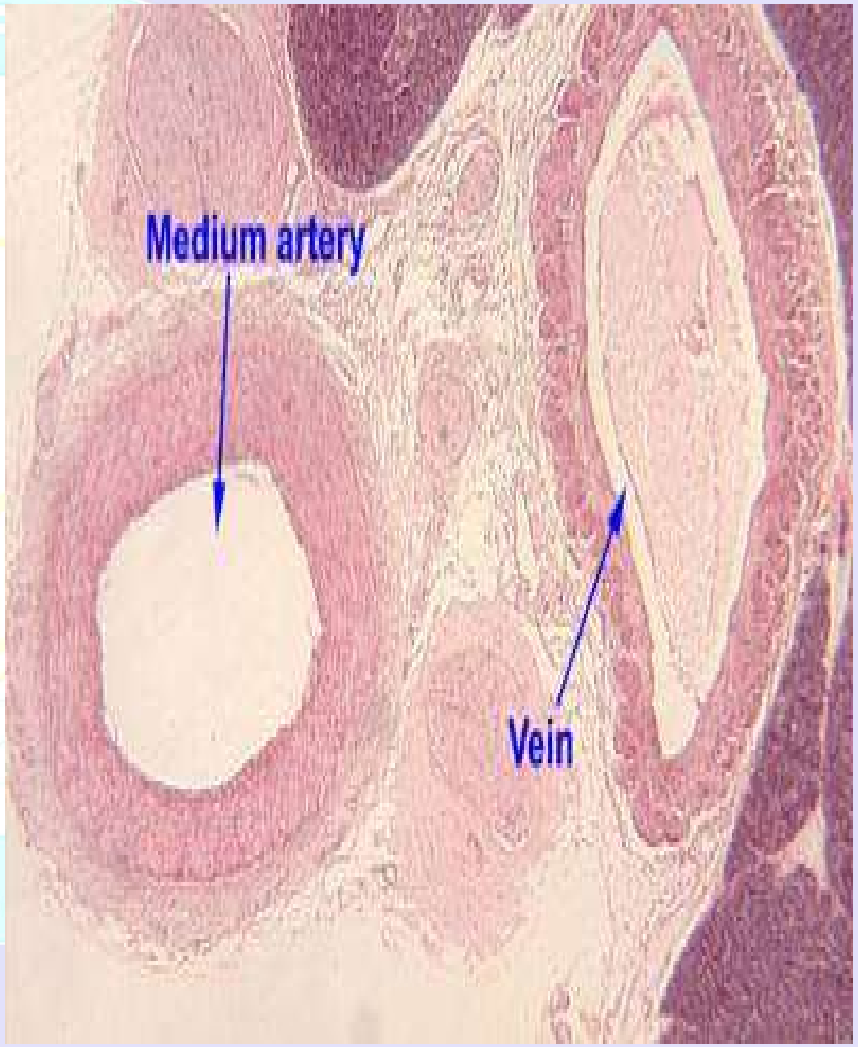
ARTERIOLES AND SMALL ARTERIES

- Due to narrow lumina and relatively thick walls they function to regulate the distribution of blood to different capillary networks by vasoconstriction or vasodilation.
- Chief constrictor of systemic blood pressure.



MEDIUM SIZED ARTERIES

- Also known as muscular arteries because of thick wall.
- Also known as distributing vessels.
- Most of the named arteries of the body belong to this group example: axillary, radial, femoral and tibial arteries.



MEDIUM SIZED ARTERIES

- The tunica intima shows all three types of layers:
  - The endothelium.
  - The subendothelium.
  - Internal elastic lamina.
- Tunica media is made up of several layers of circularly arranged smooth muscle fibers.
- Connective tissue composed mainly of collagenous fibers is found between these smooth muscle fibers.
- The tunica adventitia consists of connective tissue.

LARGE ARTERIES

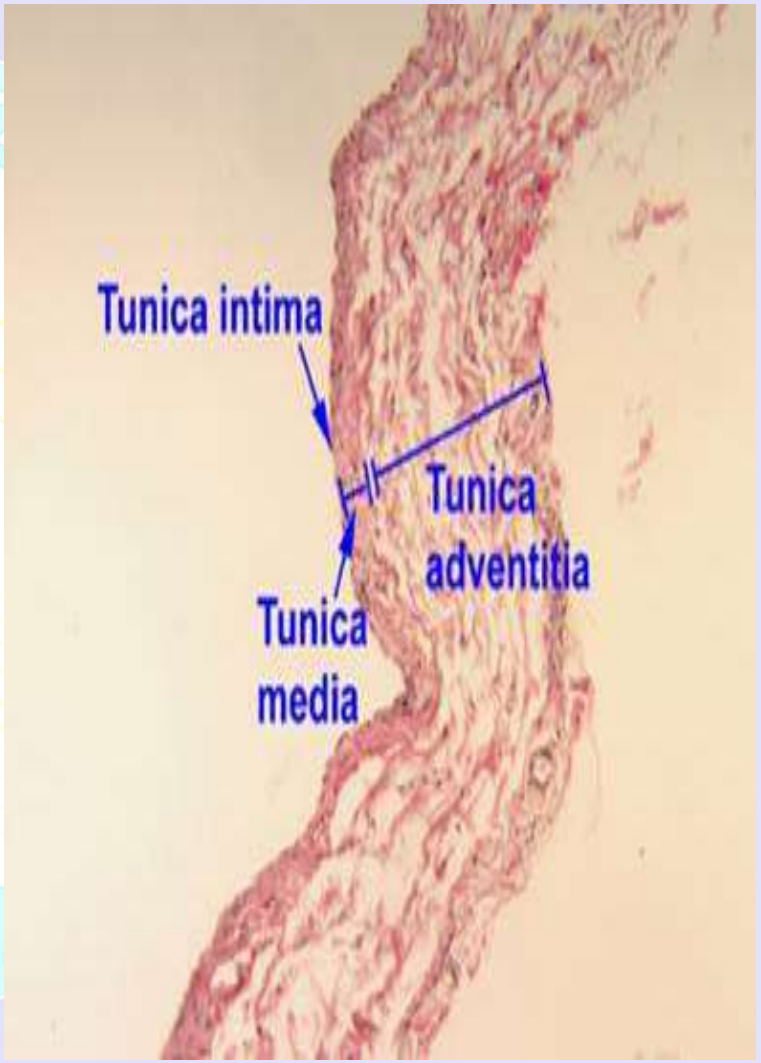
- Walls of the vessels are relatively thin in proportion to their large diameter.
- Called conducting arteries because they conduct blood from the heart to the medium sized distributing arteries.
- The intima is relatively thick and lined by endothelial cells.
- The subendothelial layer of connective tissue consists of collagenous and elastic fibers .
- The media consists of elastic fibers & series of concentrically arranged perforated elastic laminae which increase in number with age.
- The adventitia is thin and consist of collagenous fibers arranged in longitudinal spirals.

VEINS

Veins are classified into three groups:

- Venules.
- Medium sized veins.
- Large veins

- The walls of veins are thinner than the walls of arteries, while their diameter is larger.
- In contrast to arteries, the layering in the wall of veins is not very distinct.

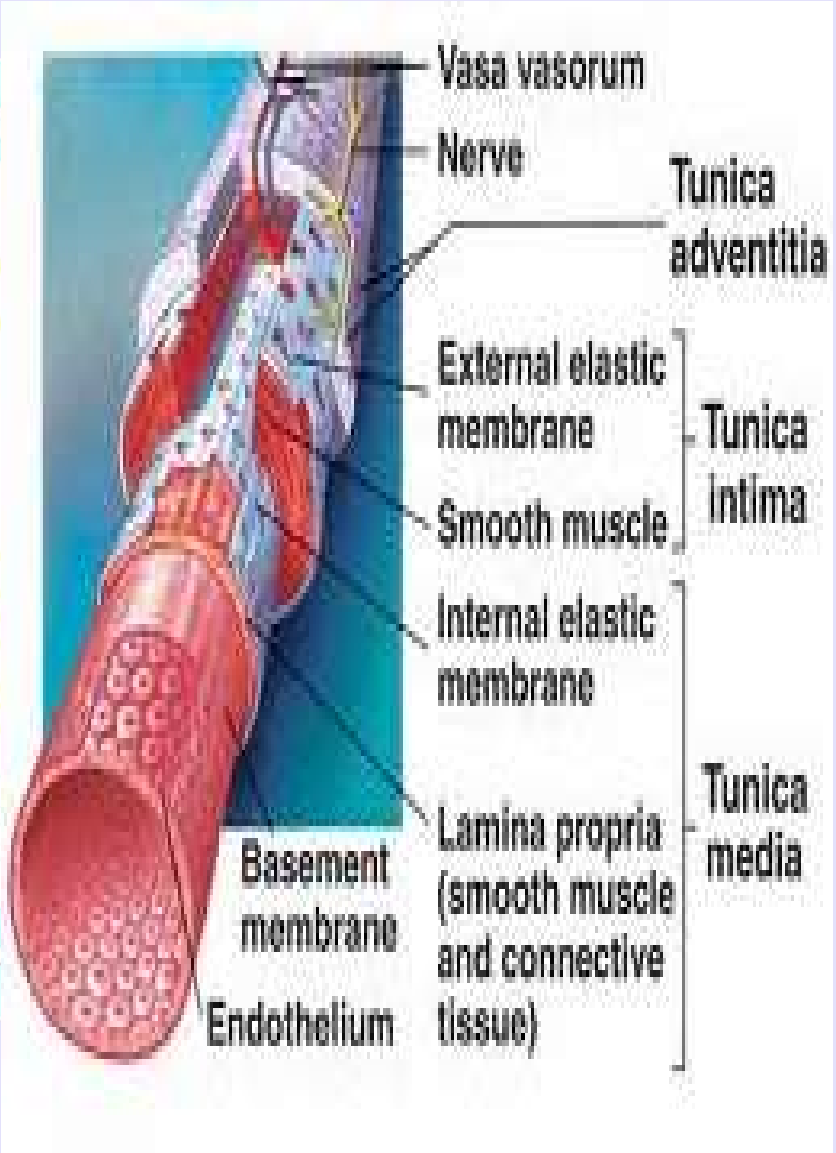


VEINS

- The tunica intima is very thin.
- Only the largest veins contain an appreciable amount of subendothelial connective tissue.
- Internal and external elastic laminae are absent or very thin.
- The tunica media appears thinner than the tunica adventitia, and the two layers tend to blend into each other.
- The appearance of the wall depends on the location.
- The walls of veins in the lower parts of the body are typically thicker than those of the upper parts of the body.
- The walls of veins which are embedded in tissues that may provide some structural support are thinner than the walls of unsupported veins.

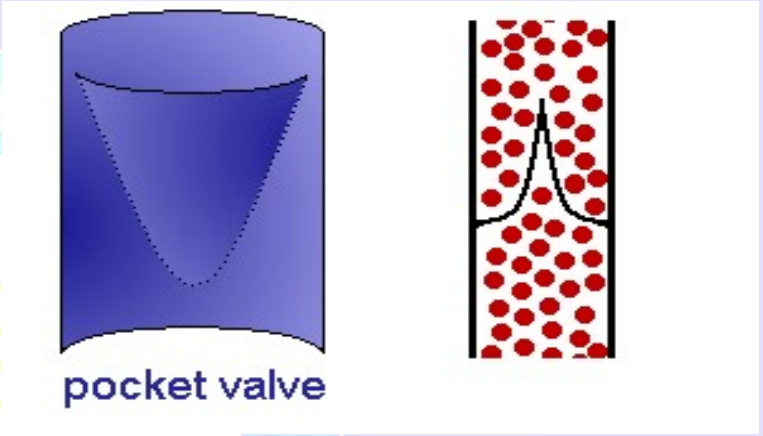
VENULES

- They are larger than capillaries. Small venules are surrounded by pericytes.
- A few smooth muscle cells may surround larger venules.
- The venules merge to form small to medium-sized veins



SMALL TO MEDIUM-SIZED VEINS

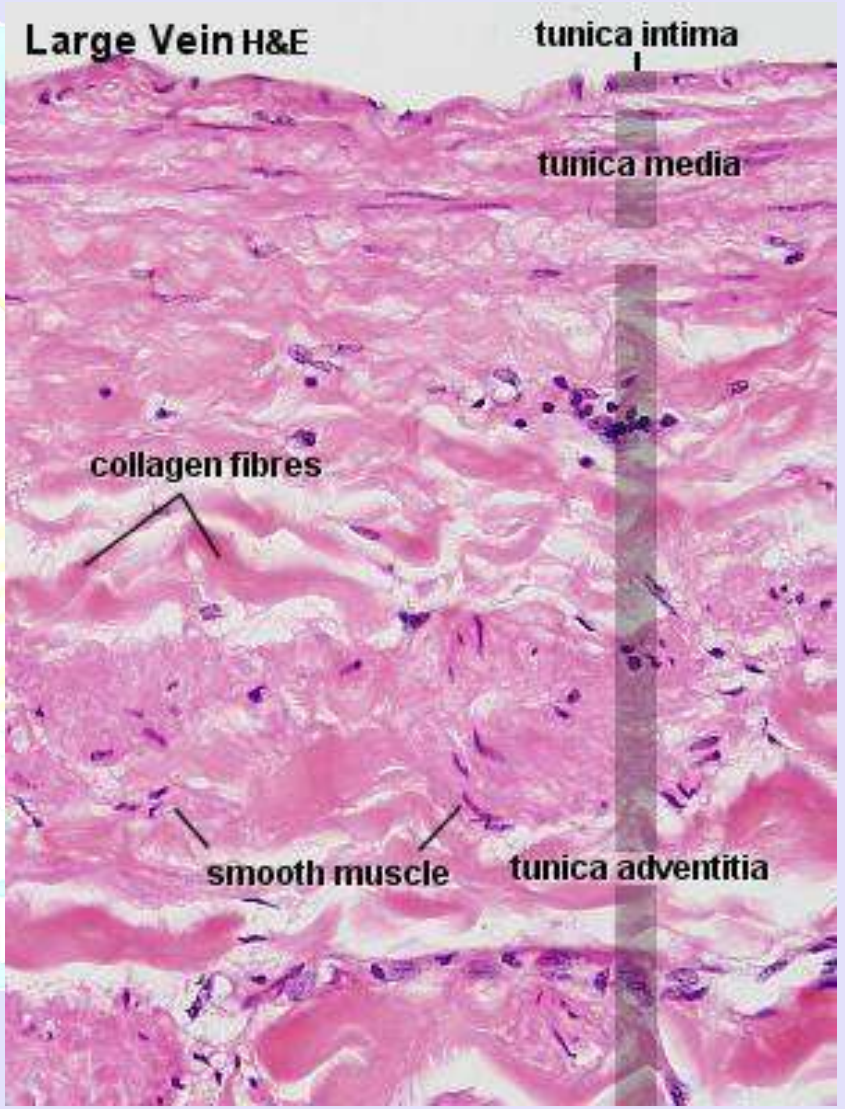
- Contain bands of smooth muscle in the tunica media. The tunica adventitia is well developed.
- Tunica adventitia contains longitudinally oriented bundles of smooth muscle.
- Small to medium-sized veins are characterised by the presence of valves.
- The valves are formed by loose, pocket-shaped folds of the tunica intima, which extend into the lumen of the vein





THE LARGE VEINS

- Contain subendothelial connective tissue in the tunica intima, but both intma and the tunica media are still comparatively thin.
- Collagen and elastic fibers are present in the tunica media.
- The tunica adventitia is very wide, and it contains bundles of longitudinal smooth muscle.
- The transition from the tunica adventitia to the surrounding connective tissue is gradual.
- Valves are absent.

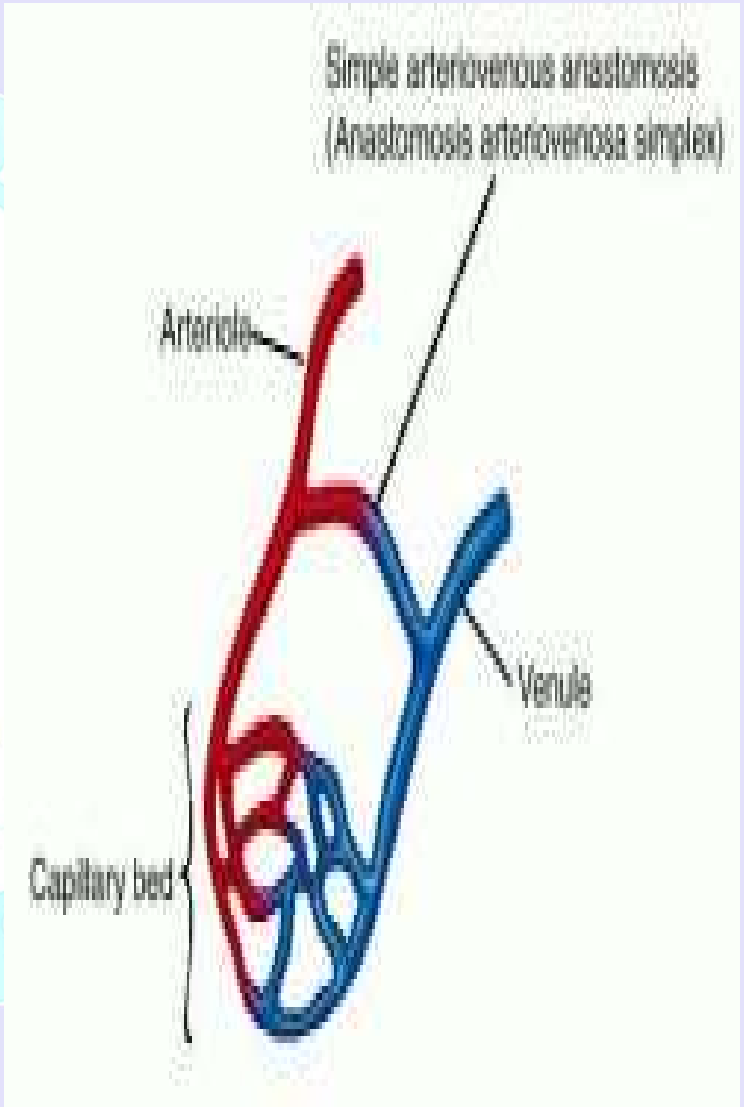


ANASTOMOSIS

- Connection of two structures. It refers to connections between blood vessels.

- The circulatory anastomosis is further divided into arterial and venous anastomosis.

- Arterial anastomosis includes actual arterial anastomosis (eg. Palmar arch, plantar arch) and potential arterial anastomosis (eg. Coronary arteries and cortical branch of cerebral arteries)



REFERENCES

- **BASIC HISTOLOGY**

JUNQUEIRA, LUIZ CARLOS,LANGE.

10TH EDITION,

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THANK YOU